Revised A1-C: service work and causal traffic state

15 October 2026 — conditional module, diagnostic and new choice

Status. Revised A1-C's qualification direction is user-approved; M2-A1 is closed at that scope. The equations below are proposed/diagnostic, not an approved complete M2 traffic law. Stop at A1-C-S1 before selecting the state/speed dynamics. No simulator or traffic experiment is involved.

1. One work account and a concrete allocation module

Retain the finite terminal histories, common advice, triangular backbone and physical/eligible stocks:

$$P_j(t) = F_j(t - b/v), \quad q_j = P_j - Y_j, \quad N_j = F_j - Y_j, \quad C(v) = \frac{wk_j v}{w + v}, \qquad j = A, E, B. \tag{1}$$

As before, eligible demand is $d_j = C(v)$ for $q_j > 0$, and $d_j = \dot{P}_j$ at an empty eligible queue. All lane offers are at most $C(v)$. A causal solution must maintain $Y_j \le P_j$.

Let X denote the justified current aggregate traffic state and Z the retained service/recovery state. For fixed inputs, consider

$$0 \le B \le R \le C(v_-), \qquad c_A(X, Z), c_E(X, Z) \ge 1, \qquad c_A g_A + c_E g_E \le B. \tag{2}$$

B is available normalized service work per second; g_j is admitted vehicle-equivalent mass per second; c_j is work per unit mass relative to ordinary passage. Only when $c_j = 1$ do work and mass rates coincide. There is one budget, not a separate recovery constraint per past entry. The complete model must predict B from its account, not receive a future service trace from outside. Here it is an input only to the local module. The maps B, c_j and the physical division of current versus retained burden remain to be qualified; they cannot both charge the same effect.

Proposed fair-work convention. For these fixed inputs choose the smallest $z \ge 0$ with

$$H(z) := \min\{c_A d_A, z\} + \min\{c_E d_E, z\} = \min\{B, c_A d_A + c_E d_E\}, \quad g_j = \min\{d_j, z/c_j\}. \tag{3}$$

If both offers are zero, take $z = 0$. H is continuous and strictly increasing until all offered work is served; hence (3) determines unique flows. If work demand does not exceed B, both offers pass. Otherwise the full work budget is used, with unused shares reallocated. In all cases

$$0 \le g_j \le d_j, \qquad g_A + g_E \le B \le R. \tag{4}$$

Equation (3) is a declared equal-work fairness assumption, not a theorem of traffic behavior. Conservative allocation templates motivate its shape; the source literature does not establish our state-to-work maps.

2. Shares, effective throughput and residual service

When both streams can use their work shares,

$$z = B/2, \qquad g_A = \frac{B}{2c_A}, \quad g_E = \frac{B}{2c_E}, \quad s_E = \frac{g_E}{g_A + g_E} = \frac{c_A}{c_A + c_E}. \tag{5}$$

Thus $c_E > c_A$ gives E a smaller mass share and also changes total flow. For $c_A = c_E = c$, the shares are equal but total service is B/c. For $c_E = c_A(1 + \varepsilon)$, E's share is $1/(2 + \varepsilon)$. These are conditional implications, not an approved equality or small ε assumption for real lane-drop traffic.

For an actual or specified offered mix θ_j summing to one,

$$\mu_{\text{mix}} = \frac{B}{\theta_A c_A + \theta_E c_E}. \tag{6}$$

At binding work demand, actual total outflow equals the reading for its admitted mix; at underload it may be smaller. Do not infer capacity loss from low outflow. A composition-independent capacity is not asserted by (6), and no mix is inferred from absent traffic. This is one shared account with a composition-dependent mass conversion.

If A's offer is served fully and E remains work-limited,

$$g_A = d_A, \qquad g_E = \frac{B - c_A d_A}{c_E}, \qquad c_A d_A \le B/2, \quad c_E d_E \ge B - c_A d_A. \tag{7}$$

A larger E cost can reduce use of nominal spare mass service, unlike a priority weight that only redistributes a fixed mass budget.

B	c_A	c_E	(g_A, g_E) when both ready	total
6	1	1	$(3, 3)$	6
6	1	3/2	$(3, 2)$	5
6	1	2	$(3, 3/2)$	9/2
6	2	2	$(3/2, 3/2)$	3

These normalized exact hand allocations do not assign physical traffic parameters. For $B = 6, d_A = 2, c_A = 1, c_E = 2$ and ample E offer, (7) gives $(g_A, g_E) = (2, 2)$. At underload, e.g. $d_A = d_E = 1$ with these costs, both pass and unused work is positive.

Not yet a full model. Fixed-input uniqueness does not establish uniqueness after costs depend on endogenous speed, queues or recovery. The next page tests one tempting instantaneous closure. No numerical smoothing, clipping epsilon or timestep-selected branch is introduced.

3. An exact queue-birth ambiguity in a simple completion

Use normalized $v = w = 1, k_j = 2$, so $C(v) = 1$ and the congested FD speed is $u_c(f) = f/(2-f)$. Over a local time interval take

$$B = R = 4/5, \quad \dot{P}_A = a = 1/5, \quad \dot{P}_E = \lambda = 1/2, \quad q_A = 0, \quad q_E(0) = 0, \quad c_A = 1. \tag{8}$$

The combined ordinary offer is $7/10 < 4/5$. Illustrate a monotone speed-cost law and a queue-selected speed by

$$c_E = 2 - u_E, \qquad u_E = \begin{cases} 1, & q_E = 0, \\ g_E/(2-g_E), & q_E > 0. \end{cases} \tag{9}$$

This cost law is a diagnostic assumption, not an empirical formula or the selected A1-C closure. The recovery budget is held fixed to isolate the instantaneous entry-state test. This local constant-offer interval is not asserted to be an exact H1-generated finite-cohort MP6 witness.

At $q_E = 0$, $c_E = 1$ and work demand $a + \lambda = 7/10 < B$, so $g_A = a, g_E = \lambda$ and $\dot{q}_E = 0$. At $q_E > 0$, E can send at 1 and A uses its full offer. Equation (7) requires

$$g_E \left(2 - \frac{g_E}{2-g_E}\right) = \frac{3}{5} \quad \Longleftrightarrow \quad 15g_E^2 - 23g_E + 6 = 0. \tag{10}$$

The roots are $1/3$ and $6/5$; only $1/3$ is feasible. Then $u_E = 1/5, c_E = 9/5, g_A = 1/5$, the work budget is exactly $4/5$, and

$$\dot{q}_E = \lambda - g_E = 1/6 \quad (q_E > 0), \qquad \dot{q}_E = 0 \quad (q_E = 0). \tag{11}$$

For every chosen $t_b \geq 0$, as well as $t_b = \infty$, the function

$$q_{t_b}(t) = \frac{[t - t_b]_+}{6}, \quad Y_E(t) = t/2 - q_{t_b}(t), \quad Y_A(t) = t/5 \tag{12}$$

satisfies the absolutely-continuous queue/departure rules almost everywhere on the local offer interval. Both departures are nonnegative, increasing, eligible and budget-feasible. Yet queue birth is arbitrary.

The differential equation holds outside isolated transition times and does not choose the next regime there. An explicit rule for continuation could exclude (12), but it would be an additional scientific rule. A discrete integrator selecting a branch does not qualify that rule. This example concerns the stated closure, not all A1-C laws or a complete P-H history model.

4. Conditional physical and clearance interfaces

If a complete causal law supplies admissible departures, retain

$$H = k_j b > M, \qquad F_j \le M, \qquad 0 \le \dot{F}_j \le C(v). \tag{13}$$

For the initially empty terminal, vacancy before b/w is positive since $F_j(t) \le C(v)t < k_j w t$. Afterwards

$$V_j(t) = Y_j(t - b/w) + H - F_j(t) \ge H - M > 0. \tag{14}$$

Thus the sufficient domain protects the H1 handoff under either future state closure. This is not a general upstream spillback solver.

Physical stock is $N_j = F_j - Y_j$, never service memory or weighted mass. For uniform initial footprint $[0, \ell]$ and road length L, the complete local cohort objective remains

$$J(v) = \frac{M(L - \ell/2)}{v} + \int_0^\infty (q_A + q_E + q_B)\, dt. \tag{15}$$

Actual departure mass is $g_j dt$, not $c_j g_j dt$. A completed entry may retain service influence while contributing no further local vehicle-time. B retains its independent admitted domain and $Y_B = P_B$.

A sufficient clearance condition only. All traffic is eligible by $t_* = L/v$, and B is clear then. Suppose the future qualified account has uniform bounds

$$1 \le c_j(t) \le c_{\max} < \infty, \quad 0 \le B(t) \le R, \quad \int_{t_*}^\infty (R - B(t))dt \le \Lambda < \infty. \tag{16}$$

If $Q = q_A + q_E > 0$, one lane offers $C(v) \ge R$, hence its work offer reaches at least B. The account binds and $B = c_A g_A + c_E g_E \le c_{\max}(g_A + g_E)$. Before clearance,

$$Q(T) \le Q(t_*) - \frac{R(T - t_*) - \Lambda}{c_{\max}}. \tag{17}$$

Therefore clearance occurs by

$$t_* + \frac{c_{\max} Q(t_*) + \Lambda}{R}. \tag{18}$$

Temporary zero budget is allowed; no artificial positive floor is assumed. The complete M2 law must establish (16) or a different valid clearance argument. The work module alone and M1's finite-kernel proof do not do so.

Information and gradients. Known initialization plus a uniquely autonomous state law could preserve ending-statistic reconstruction. That is conditional: no full-law uniqueness, consequential reachable MP6 regime, observation collision, action split or useful gradient is proved. Changes in cost, state and event times all enter the full-objective path; none may be frozen merely to simplify a later derivative check.

5. What must be selected before exact-law qualification

S-E: equilibrium boundary state plus explicit causal transitions. The free branch persists while its physical admission/service conditions permit it. A specified restriction must trigger queue formation; a specified clearance condition returns the stream to free passage. This rejects the arbitrary births in (12) if the free conditions remain satisfied. It does not supply a complete entropy/transition rule or prove the whole history node unique. A simple speed-only cost can remain benign under ordinary underload; conditional compulsory entry/opportunity must still be modeled.

S-V: finite-rate local aggregate speed state. Predict approach/entry speeds from explicit local initial-value dynamics and admission/empty-state rules. Queue presence is a constraint/input, not an instantaneous complete speed assignment. This changes local motion/state meaning and needs parameter and averaging semantics. It is not automatically microscopic car following or a spatial CTM model, and an added speed ODE alone does not guarantee a valid node.

Both choices may use a unified service map that has intermediate states; neither requires linear or globally smooth interpolation. They differ in how the traffic state changes through time, not in whether forced interaction and ordinary congestion may coexist. Neither is selected. The provisional first qualification is S-E because it retains the smaller first-order amendment; its failure would warrant an explicit return to S-V.

Sources and adaptation. The boundary formulation in Jin et al. (sections 2 and 6) has its own selection rule and proof. The inspected relation in Leclercq et al. (section 3.2) assumes congested entry. Neither paper supplies this work node. The fairness convention, state-to-cost proposal and queue-birth diagnostic are review constructions, not proofs of the finite task.

Gate and original-stage status. A1-C-S1 is Type 2 G3; select the local state/dynamics interpretation before qualifying its exact equations, recovery/overlap and joint MP6 domain. M2-A1's revised A1-C direction is closed. H1/cutoff, known initial information, finite storage, primary ordinary-underload domain, local choke completion and S-P–S-L remain. The approved three-part review and Step 3 are unfinished. No simulator, installation, calibration, traffic experiment or numerical gradient has been undertaken. The strict, fixed-share, aggregate/history/scalar and search rivals remain; a common fastest action is an admissible outcome.